Report for E-commerce Business

(Data Analysis and Dashboard)

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Data Overview

- Date Period: January 1, 2023 - March 31, 2023

```
SELECT MIN(Date) as minDate, MAX(Date) AS maxDate FROM `e-commerce-449804.raw_data.e_commerce_raw`
```

Query results

JOB INFORMATION | RESULTS | CHART | JSON | EXECUTION DETAILS | EXECUTION GRAPH

Row	minDate	maxDate
1	2023-01-01	2023-03-31

- Traffic Sources: 5 distinct sources (Direct, Email Campaigns, Google Ads, Organic Search, Social Media)

```
SELECT DISTINCT Traffic_Source FROM `e-commerce-449804.raw_data.e_commerce_raw`
ORDER BY Traffic_Source ASC
```

Query results

JOB INFORMATION | RESULTS | CHART | JSON | EXECUTION DETAILS | E

Row	Traffic_Source
1	Direct
2	Email Campaign
3	Google Ads
4	Organic Search
5	Social Media

- Marketing Campaigns: 5 campaigns (Flash Deals, Holiday Offers, Summer Sale, Weekend Discount, Winter Promo)

```
SELECT DISTINCT Campaign FROM `e-commerce-449804.raw_data.e_commerce_raw`
ORDER BY Campaign ASC
```

Query results

JOB INFORMATION | RESULTS | CHART | JSON | EXECUTION DETAILS

Row	Campaign
1	Flash Deals
2	Holiday Offers
3	Summer Sale
4	Weekend Discount
5	Winter Promo

- Customer Types: New and Returning

```sql
SELECT DISTINCT Customer_Type FROM `e-commerce-449804.raw_data.e_commerce_raw`
ORDER BY Customer_Type ASC
```

Query results

JOB INFORMATION | RESULTS | CHART | JSON | EXECUTION DETAILS | EXE

Row	Customer_Type
1	New
2	Returning

- Total Visitors: 492,888 (Sum of Visitors)

```sql
SELECT SUM(Visitors) FROM `e-commerce-449804.raw_data.e_commerce_raw`
```

Query results

JOB INFORMATION | RESULTS | CHART | JSON | EXECUTION DETAILS

Row	f0_
1	492888

- Total Marketing Spend: $1,144,779.82

```sql
SELECT SUM(Marketing_Spend) FROM `e-commerce-449804.raw_data.e_commerce_transformed`
```

Query results

JOB INFORMATION | RESULTS | CHART | JSON | EXECUTION DETAILS | EXECUTION GR

Row	f0_
1	1144779.800000...

- Total Purchases: 71,164

```sql
SELECT SUM(Purchases) FROM `e-commerce-449804.raw_data.e_commerce_transformed`
```

Query results

JOB INFORMATION | RESULTS | CHART | JSON | EXECUTION DETAILS | EXE

Row	f0_
1	71164.0

- Total Revenue: $2,436,900.63

```
SELECT SUM(Revenue) FROM `e-commerce-449804.raw_data.e_commerce_transformed`
```

Query results

JOB INFORMATION | RESULTS | CHART | JSON | EXECUTION DETAILS

Row	f0_
1	2436900.6333333314

- Cart Abandonment Rate (Average): 29.53%

```
SELECT AVG(Cart_Abandonment_Rate) FROM `e-commerce-449804.raw_data.e_commerce_raw`
```

Query results

JOB INFORMATION | RESULTS | CHART | JSON | EXECUTION DETAILS | EXECUT

Row	f0_
1	29.528866666666683

- Repeat Purchase Rate (Average): 17.65%

```
SELECT AVG(Repeat_Purchase_Rate) FROM `e-commerce-449804.raw_data.e_commerce_raw`
```

Query results

JOB INFORMATION | RESULTS | CHART | JSON | EXECUTION DETAILS | EXECUTIO

Row	f0_
1	17.65037777777...

- Conversion Rate (Average): 5.64%

```
SELECT AVG(Conversion_Rate) FROM `e-commerce-449804.raw_data.e_commerce_raw`
```

Query results

JOB INFORMATION | RESULTS | CHART | JSON | EXECUTION DETAILS

Row	f0_
1	5.640599999999...

Data Preprocess and Cleaning

Step1: Checked data type on each column.

e_commerce_raw QUERY SHARE COPY

SCHEMA DETAILS PREVIEW TABLE EXPLORER PREVIEW

Filter Enter property name or value

Field name	Type	Mode	Key
Date	DATE	NULLABLE	-
Traffic_Source	STRING	NULLABLE	-
Campaign	STRING	NULLABLE	-
Customer_Type	STRING	NULLABLE	-
Visitors	INTEGER	NULLABLE	-
Marketing_Spend	FLOAT	NULLABLE	-
Add_to_Cart	INTEGER	NULLABLE	-
Purchases	FLOAT	NULLABLE	-
Revenue	FLOAT	NULLABLE	-
Cart_Abandonment_Rate	FLOAT	NULLABLE	-
Repeat_Purchase_Rate	FLOAT	NULLABLE	-
Conversion_Rate	FLOAT	NULLABLE	-

Step2: Checked for null values in each data attribute and found the following attributes with null values:

1. Marketing_Spend
2. Purchases
3. Revenue

Step3: Replaced the null values with the **mean** for each data point and created a transformed view based on the raw data table.

e_commerce_transformed RUN SAVE VIEW OPEN IN MORE This query will process 46.25 KB when run.

```
SELECT
Date
,Traffic_Source
,Campaign
,Customer_Type
,Visitors
,COALESCE(Marketing_Spend, (SELECT AVG(Marketing_Spend) FROM `e-commerce-449804.raw_data.e_commerce_raw`)) AS Marketing_Spend
,Add_to_Cart
,ROUND(COALESCE(Purchases, (SELECT AVG(Purchases) FROM `e-commerce-449804.raw_data.e_commerce_raw`))) AS Purchases
,COALESCE(Revenue, (SELECT AVG(Revenue) FROM `e-commerce-449804.raw_data.e_commerce_raw`)) AS Revenue
,Cart_Abandonment_Rate
,Repeat_Purchase_Rate
,Conversion_Rate
FROM `e-commerce-449804.raw_data.e_commerce_raw`
```

Press Alt+F1 for Accessibility Options.

Query results SAVE RESULTS OPEN IN

JOB INFORMATION RESULTS CHART JSON EXECUTION DETAILS EXECUTION GRAPH

Row	Date	Traffic_Source	Campaign	Customer_Type	Visitors	Marketing_Spend	Add_to_Cart	Purchases	Revenue	Cart_Abandonment_
1	2023-01-01	Email Campaign	Flash Deals	New	1226	1469.85	261	46.0	6416.7	19.11
2	2023-01-01	Social Media	Flash Deals	New	1510	505.47	133	183.0	1180.24	28.07
3	2023-01-02	Social Media	Flash Deals	New	1559	4429.12	274	153.0	6692.86	49.04
4	2023-01-04	Direct	Flash Deals	New	1394	4770.05	162	187.0	5613.54	10.42
5	2023-01-05	Email Campaign	Flash Deals	Returning	1222	1844.36	210	252.0	6901.51	12.74
6	2023-01-06	Email Campaign	Flash Deals	New	1195	2808.55	96	287.0	1622.2	20.43
7	2023-01-06	Google Ads	Flash Deals	Returning	1600	4899.6	112	117.0	4137.16	14.99
8	2023-01-06	Social Media	Flash Deals	New	1138	3711.47	259	245.0	5636.58	41.49
9	2023-01-06	Google Ads	Flash Deals	New	1204	2543.955111111...	391	52.0	1836.36	17.47
10	2023-01-06	Social Media	Flash Deals	New	1782	4481.66	200	96.0	8632.14	40.53
11	2023-01-07	Direct	Flash Deals	New	798	2453.81	112	153.0	3313.43	13.76

Data Analysis

This report analyzes website traffic, marketing spend, and sales performance over the last three months.

Key findings include:

1. Sales and Revenue

- There are 150,000 to 172,000 visitors, generating over $700,000 in revenue from more than 20,000 purchases per month.

```sql
SELECT
DATE_TRUNC(Date, month) AS yearMonth,
SUM(Visitors) AS totalVisitors,
ROUND(SUM(Revenue),2) AS totalRevenue,
SUM(Purchases) AS TotalPurchases
FROM `e-commerce-449804.raw_data.e_commerce_transformed`
GROUP BY yearMonth
```

Query results

JOB INFORMATION | RESULTS | CHART | JSON | EXECUTION DETAILS

Row	yearMonth	totalVisitors	totalRevenue	TotalPurchases
1	2023-01-01	171183	827517.95	24470.21234567...
2	2023-02-01	154550	778166.66	20891.89629629...
3	2023-03-01	167155	831216.03	25809.00246913...

- Total purchases reached 71,164, with a total revenue of $2,436,900.63.
- Based on the above metrics, the revenue per visitor is approximately $5.00. The code and the results are provided below.

Untitled query RUN SAVE DOWNLOAD SH

```sql
WITH
  main AS (
  SELECT
    DATE_TRUNC(Date, month) AS yearMonth
    ,SUM(Visitors) AS totalVisitors
    ,ROUND(SUM(Revenue),2) AS totalRevenue
    ,SUM(Purchases) AS totalPurchases
  FROM
    `e-commerce-449804.raw_data.e_commerce_transformed`
  GROUP BY
    yearMonth
    )
SELECT
  yearMonth
  ,ROUND(totalRevenue/totalVisitors,2) AS revenuePerVisitor
FROM
  main
ORDER BY yearMonth ASC
```

Query results

JOB INFORMATION | RESULTS | CHART | JSON | EXECUTION

Row	yearMonth	revenuePerVisitor
1	2023-01-01	4.83
2	2023-02-01	5.04
3	2023-03-01	4.97

2. Traffic Breakdown

- From the all time data, the majority of visitors have come from **Direct**, while **Organic Search** has generated the fewest visits. (Note: The number of visitors is pre-calculated, and we cannot obtain unique visitor data with the data provided.)

Untitled query

RUN SAVE DOWNLOAD

```sql
WITH
  main AS (
  SELECT
    Traffic_Source
    ,SUM(Visitors) AS totalVisitors
  FROM
    `e-commerce-449804.raw_data.e_commerce_raw`
  GROUP BY
    Traffic_Source
    )
SELECT
  Traffic_Source
  ,totalVisitors
FROM
  main
ORDER BY totalVisitors DESC
```

Query results

JOB INFORMATION | RESULTS | CHART | JSON | EXEC

Row	Traffic_Source	totalVisitors
1	Direct	114651
2	Google Ads	106263
3	Social Media	98681
4	Email Campaign	89993
5	Organic Search	83300

- Conversion rates were highest from **Email Campaigns** (5.99%).

```sql
WITH
  main AS (
  SELECT
    Traffic_Source
    ,AVG(Conversion_Rate) AS avgConversion_Rate
  FROM
    `e-commerce-449804.raw_data.e_commerce_transformed`
  GROUP BY
    Traffic_Source
    )
SELECT
  Traffic_Source
  ,avgConversion_Rate
FROM
  main
ORDER BY avgConversion_Rate DESC
```

Query results

JOB INFORMATION | RESULTS | CHART | JSON | EXECU

Row	Traffic_Source	avgConversion_Rate
1	Email Campaign	5.994137931034...
2	Google Ads	5.838947368421...
3	Social Media	5.661511627906...
4	Organic Search	5.425316455696...
5	Direct	5.306699029126...

3. Campaign Behavior

- The **Email Campaign** has the highest average **conversion rate, at 6%** when considering all time data.

Untitled query

RUN SAVE DOWNLOAD

```sql
WITH
  main AS (
  SELECT
    Traffic_Source
    ,AVG(Conversion_Rate) AS avgCR
  FROM
    `e-commerce-449804.raw_data.e_commerce_raw`
  GROUP BY
    Traffic_Source
    )
SELECT
  Traffic_Source
  ,avgCR
FROM
  main
ORDER BY avgCR DESC
```

Query results

JOB INFORMATION | RESULTS | CHART | JSON | EXEC

Row	Traffic_Source	avgCR
1	Email Campaign	5.9941379310344844
2	Google Ads	5.8389473684210538
3	Social Media	5.6615116279069779
4	Organic Search	5.4253164556962012
5	Direct	5.3066990291262126

4. Marketing Spend Breakdown

- The **Winter Promo** campaign has consumed the majority of the marketing budget over time. ($242,248.52)
- The **Holiday Offers** campaign had the highest all-time conversion rate at **5.96%**, followed by the **Winter Promo** with the second highest at **5.73%**.

```sql
WITH
  main AS (
  SELECT
    Campaign
    ,SUM(Marketing_Spend) AS Marketing_Spend
    ,AVG(Conversion_Rate) AS avgCR

  FROM
    `e-commerce-449804.raw_data.e_commerce_transformed`
  GROUP BY
    Campaign
    )
SELECT
  Campaign
  ,ROUND(Marketing_Spend,2) AS Marketing_Spend
  ,avgCR
FROM
  main
ORDER BY Marketing_Spend DESC
```

Query results

JOB INFORMATION | RESULTS | CHART | JSON | EXECUTIO

Row	Campaign	Marketing_Spend	avgCR
1	Winter Promo	242248.52	5.732857142857...
2	Flash Deals	236715.04	5.606853932584...
3	Weekend Discount	227049.6	5.390574712643...
4	Summer Sale	224795.03	5.509
5	Holiday Offers	213971.62	5.961046511627...

5. Customer Behavior Insights

- **New customers** had a higher average revenue per purchase and better repeat purchase rates than **Returning customers**.

```sql
WITH
  main AS (
  SELECT
    Customer_Type
    ,AVG(Revenue) AS avgRevenue
    ,AVG(Repeat_Purchase_Rate) AS avgRepeatPurchaseRate
  FROM
    `e-commerce-449804.raw_data.e_commerce_transformed`
  GROUP BY
    Customer_Type
    )
SELECT
  Customer_Type
  ,ROUND(avgRevenue,2) AS avgRevenue
  ,avgRepeatPurchaseRate
FROM
  main
ORDER BY avgRevenue DESC
```

Query results

JOB INFORMATION | RESULTS | CHART | JSON | EXECUTION DETAIL

Row	Customer_Type	avgRevenue	avgRepeatPurchaseF
1	New	5455.11	18.11685589519...
2	Returning	5374.12	17.16701357466...

- **New customers** accounted for **36,097 purchases**, while **returning customers** contributed **35,067 purchases**.
- The total number of **visitors** from **new customers is 252,398**, while **returning customers** account for **240,490**.

```sql
WITH
  main AS (
  SELECT
    Customer_Type
    ,SUM(Purchases) AS totalPurchases
    ,SUM(Visitors) AS totalVisitors

  FROM
    `e-commerce-449804.raw_data.e_commerce_transformed`
  GROUP BY
    Customer_Type
    )
SELECT
  Customer_Type
  ,totalPurchases
  ,totalVisitors
FROM
  main
ORDER BY totalPurchases DESC
```

Query results

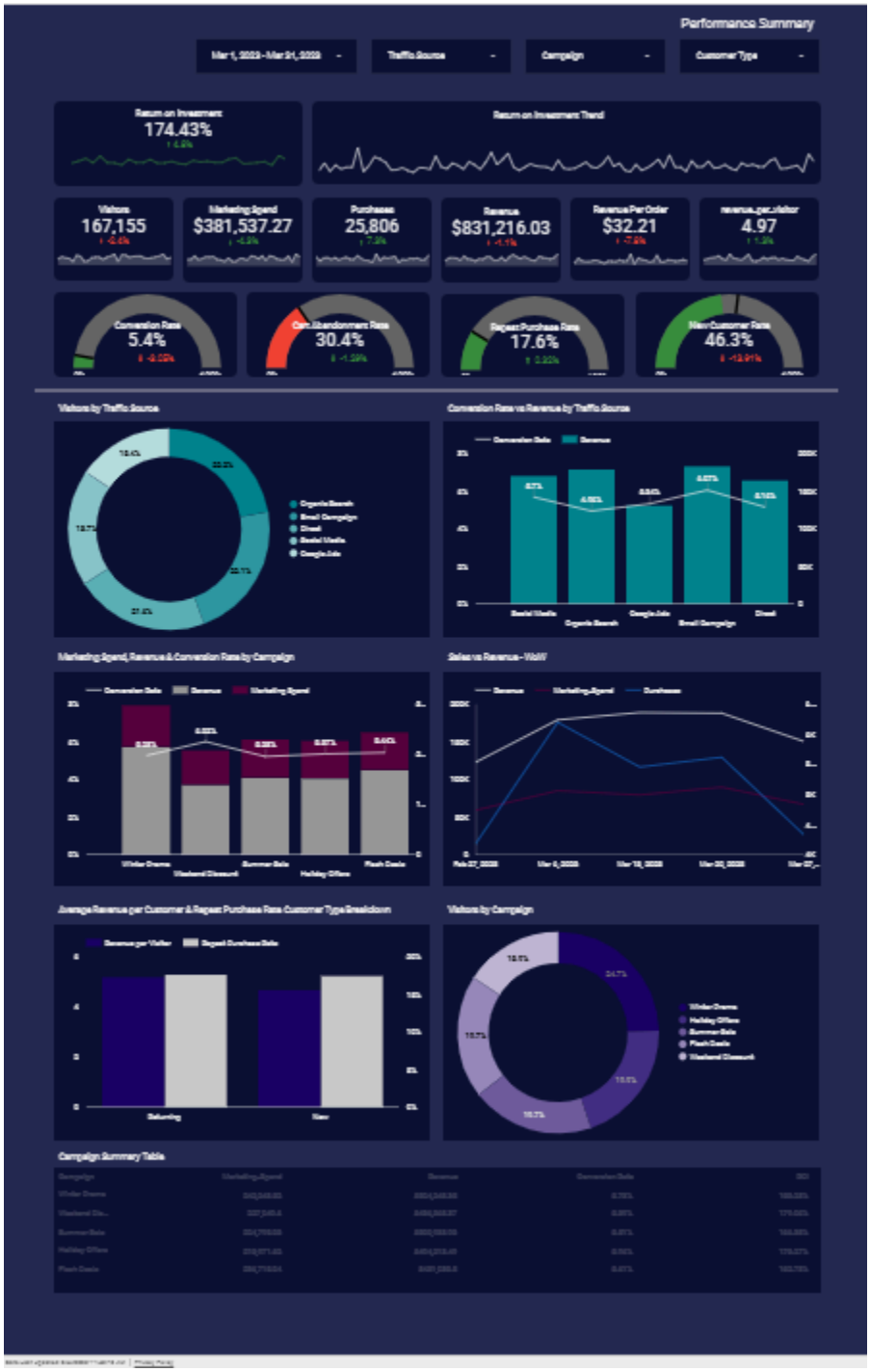
JOB INFORMATION | RESULTS | CHART | JSON | EXECUTION

Row	Customer_Type	totalPurchases	totalVisitors
1	New	36097.0	252398
2	Returning	35067.0	240490

Dashboard Overview

https://lookerstudio.google.com/reporting/41921bd5-3905-427e-ae98-c0e500de6da9/page/xrBlE

Page1: Performance Summary



Page2: Traffic Source Summary

Page3: Campaign Summary

Strategic Analysis

1. Campaign Performance
    - Strengths: Email campaigns had the highest conversion rate, indicating that targeting existing subscribers can be highly effective.
    - Weaknesses: Despite high marketing spend, the "Flash Deals" campaign had a relatively modest conversion rate (5.61%).
2. Customer Segmentation Analysis
    - New customers proved to be more valuable, with higher average revenue per visitor and better repeat purchase rates.
3. Conversion Rate Assessment
    - Direct traffic had the highest conversion rate and the total number of visitors, modest conversion rate (5.31%).
    - Bottlenecks likely exist within the traffic source, particularly from email campaign traffic sources.
4. Cart Abandonment and Retention Rates
    - Cart abandonment rates were highest for organic search traffic (31.06%) and email campaigns (30.89%).

Recommendations

1. Increase Conversion Rates
    - Focus on optimizing the sales funnel for organic search, which has the higher number of visitors but the lowest conversion rate.
    - A/B testing landing pages and improving the checkout process can help convert more visitors into customers.
2. Improve Marketing ROI
   Redistribute marketing spend to better-performing campaigns like weekend discount, which had higher conversion rates and a lower cost per acquisition.
   Consider reallocating a portion of the "Flash Deals" campaign budget to other channels with higher ROI.
3. Reduce Cart Abandonment and Boost Customer Retention
   Implement retargeting campaigns and cart recovery emails to reduce cart abandonment rates, especially from organic search and email campaign sources.
   Strengthen loyalty programs and personalized offers for returning customers to boost repeat purchase rates.